



CSCI 6344 Intro to Data Science Syllabus

Course Information

WED 06:30 PM 09:00 PM Edinburg EIEAB 1.208

Traditional Face-to-Face

Instructor Information

Marzieh Ayati

Email: marzieh.ayati@utrgv.edu

Office Location

EIEAB 3.217

Office Hours: By Appointment and Wed

Course Description

This course provides an introduction to computational and statistical approaches to analyzing data. This Data Science course is designed to provide a comprehensive introduction to the field of data science. The course covers a wide range of topics including data and distributions, statistical inference, assessing pairwise relationships, and machine learning. In the first part of the course, students will learn about the basics of data and distributions, including measures of central tendency and variability, probability distributions, and statistical tests. They will also learn how to visualize data using plots and charts, and how to use statistical software to analyze data. The second part of the course focuses on statistical inference, including estimation and hypothesis testing. Students will learn about sampling distributions, confidence intervals,

and p-values, and how to use these concepts to make inferences about populations based on samples. In the third part of the course, students will learn about assessing pairwise relationships, including correlation and mutual information. They will learn how to use these techniques to examine the relationship between two or more variables. Finally, the course concludes with an introduction to machine learning. Students will learn about the various types of machine learning, including supervised and unsupervised learning, and will explore the use of machine learning algorithms for prediction and classification.

Credit Hours: 3

Tentative Calendar of Activities

Date	Topic	Quiz/Homework
Week 1 (August 26)	Syllabus, Data, Data and Matrix,	Q1, Q2,
Week 2 (Sep 2)	Basic Stat, Stat and Visualization	Q3 , Q4, HW1 (Due Sep 16)
Week 3 (Sep 9)	Data Distribution, Normalization	
Week 4 (Sep 16)	MLE, Hypothesis Testing	HW2(Due Sep 30)
Week 5 (Sep 23)	p-value, Comparing Distribution	
Week 6 (Sep 30)	Chi-square test, Multiple hypothesis testing	HW3 (Due Oct 7)
Week 7 (Oct 7)	Pairwise Association, Entropy	HW4 (Due Oct 21)
Week 8 (Oct 14)	Intro to Machine Learning, Clustering	
Week 9 (Oct 21)	Hierarchal Clustering, Recommendation	HW5 (Due Oct 28)
Week 10 (Oct 28)	Regression, Logistic regression	HW6 (Due Nov 4)
Week 11 (Nov 4)	Classification, Model Evaluation	HW7 (Due Nov 11)
Week 12 (Nov 11)	Visualization, Dimensionality Reduction	HW8 (Due Nov 25)

Date	Topic	Quiz/Homework
Week 13 (Nov 18)	Project Presentation	
Week 14 (Nov 25)	Project Presentation	
Week 15 (Dec 2)	Project Presentation	

Learning Objective for Core Curriculum Requirements

Student Learning Outcomes	Program Student Learning Outcomes	Major Assignment/Examination
a. Clean and reshape messy datasets b. Assess relationship between variables c. Use exploratory tools such as clustering and visualization tools to analyze data	Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	All homework. Final Project. Exam
d. Use methods such as regression, nearest neighbors, etc to build a classifier e. Apply dimensionality reduction tools such as principal component analysis.	Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.	All homework. Final Project. Exam

Student Learning Outcomes	Program Student Learning Outcomes	Major Assignment/Examination
f. Evaluate outcomes and make decision based on data.	Apply computer science theory and software development fundamentals to produce computing-based solutions.	All homework. Final Project. Exam

Course Learning Objectives

1. Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.
2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
3. Apply computer science theory and software development fundamentals to produce computing-based solutions.

Assessment of Learning

Grading Components:

Participation, Activities and Quiz: (30%). Every class might have at least two quizzes (in the middle and at the end).

Homeworks: (20%) The homeworks usually needs programming. These homeworks are comprised of problems that aim to allow the students to practice with the methods covered in class and develop ideas to manipulate these methods. Late homework submission will be accepted within 10 days. where a penalty of 10% for each late day will be applied. The homework should be submitted individually.

Exam: (15%) The final exam will cove all of the course material comprehensively. The students are allowed to use their notes and slides during the exam.

Project: (35%) There will be a single term project, in which the students will be expected to define research questions, obtain the data, perform analyses, and interpret their results. The students will work on the project in teams of three or four (pre-defined). The students are allowed to use a scripting language of their choice to perform the analyses for their project. R, Python, or Matlab are recommended due to easy access to rich visualization libraries. The project will have two phases:

- **Proposal Presentation and Feedback:** In the BrightSpace forum, the students will present their research questions, the data they are planning to use, and proposed analyses. Each student is also required to provide feedback to at least **four** other projects.
- **Final Presentation and Report:** (Last Weeks of Semester) The students will present their completed project in the class and submit a 2-page report describing their findings. The project will be graded based on: (i) coherence and completeness, (ii) vision and critical thinking displayed, and (iii) presentation of results, and (iv) responsiveness to feedback.
- **Teammate Evaluation.** The contribution of each team member will be assessed based on the form filled with teammate, GITHUB, or SLACK. Your final grade for evaluation will be calculated by the grade of the group multiply by the average score you get by teammates.

Grading Scale

Letter	Numerical Value
A	90 - 100
B	80 - 89
C	70 - 79
D	60 - 69
F	Below 60

Required Readings and Materials

None

Technology Needs

Programming

Please note that in this course programming will NOT be taught. You can use the following resources to get familiar with R, Python, or MATLAB:

1. Learning R: <https://www.w3schools.in/r/>
2. Learning Python: <https://www.w3schools.com/python/>
3. Learning MATLAB: <http://www.math.ucsd.edu/~bdriver/21d-s99/matlab-primer.html>

4. Learning GIT: <https://git-scm.com/>

University Attendance Policy

[UTRGV's attendance policy](#)

Course Attendance

Students are expected to attend all scheduled classes. [UTRGV's attendance policy](#) excuses students from attending class if they are participating in officially sponsored university activities, such as athletics, accommodation by Student Accessibility Services (SAS), observance of religious holy days, or military service.

If you are going to miss any class session or deadline for any reason you must contact me ahead of time or, in the case of emergency, as soon as possible. Depending on the reason, I will determine what options you have to make up the work, possibly with a penalty. In general, if you contact me ahead of time and/or have a good reason or emergency, we can work something out. If you come to me after the fact, options, if any, will be limited.

If you are sending an email to let me know that you will not be able to attend the class, the subject of the email should be "CSCI6344–Absence". Otherwise, it would not be considered.

Since I will drop the two lowest quiz scores in the final grade, you are allowed to miss two quizzes (one session).

University Policy on Recordings in Class

The use of classroom recordings is governed by the Federal Educational Rights and Privacy Act (FERPA), UTRGV's acceptable-use policy, and UTRGV HOP Policy STU 02-100 Student Conduct and Discipline. A recording of class sessions will be kept and stored by UTRGV, in accordance with FERPA and UTRGV policies. Your instructor will not share the recordings of your class activities outside of course participants, which include your fellow students, teaching assistants, or graduate assistants, and any guest faculty or community-based learning partners with whom we may engage during a class session. You may not share recordings outside of this course. As referenced in [UTRGV HOP Policy STU 02-100 Student Conduct and Discipline](#), doing so may result in disciplinary action. This is an instruction from university officials.

University AI Policy

At UTRGV, students are expected to use AI in ways that are honest, ethical, and consistent with course and assignment expectations. Because AI expectations may vary by course, instructor, assignment, and discipline, students should review the syllabus and assignment instructions before using AI. When expectations are unclear, students should ask their instructor before submitting work that involved AI. Students are responsible for checking the accuracy of AI-assisted work and protecting private or sensitive information.

Course Use of Artificial Intelligence (AI)

Students are encouraged to use Artificial Intelligence (AI) tools as part of their learning and problem-solving process. AI can be a valuable resource for exploring ideas, clarifying concepts, checking work, and improving productivity. However, AI should be used as a tool to support your thinking, not as a substitute for it.

Students are responsible for the accuracy, quality, and integrity of all work they submit. AI-generated outputs may contain errors, incorrect assumptions, fabricated information, or misleading explanations. Therefore, students are expected to critically evaluate and independently verify AI-generated content before incorporating it into their work.

Blindly copying and pasting AI-generated content is not acceptable.

If you submit work that uses AI, you should understand the methods, reasoning, calculations, code, or conclusions presented and be able to explain them in your own words.

Students may be asked to meet with the instructor to present, explain, or reproduce aspects of their submitted work. During such a meeting, you should be prepared to explain your approach, demonstrate your understanding of the underlying concepts, and answer questions about how you obtained your results, including how AI tools were used.

A simple rule: You may use AI to help you do the work, but you remain responsible for understanding, verifying, and defending the work you submit.

AI SHOULD NOT BE USED IN THE FINAL EXAM. USING AI IN THE FINAL EXAM WILL RESULT IN 0 FOR FINAL EXAM.

University Academic Integrity Statement

Academic integrity is fundamental in our actions, as any act of dishonesty conflicts as much with academic achievement as with the values of honesty and integrity. Violations of academic integrity constitute academic dishonesty and include, but are not limited to: [cheating, plagiarism \(including self-plagiarism\), and collusion](#); submission for credit of any work or materials that are attributable in whole or in part to another person; taking an examination for another person; any act designed to give unfair advantage to a student; or the attempt to commit such acts ([UT System Board of Regents Rules and Regulations](#), [UTRGV Handbook of Operating Procedures STU 02-100](#), and [UTRGV Academic Integrity Guidelines](#)). **All violations of Academic Dishonesty will be reported to Student Rights and Responsibilities in accordance with [HOP STU 02-100, Student Code of Conduct](#).**

Course Academic Integrity Policy

Zero-tolerance policy on plagiarism is enforced. If the homeworks or assignment found similar to the available resources online or to other students, you will get 0 for that assignment. If you copy (more than once) from the online resources or other students, you will get 0 for the rest of homeworks and labs until the end of semester. Although you are encouraged to work in team and help each other, the final coding and solution should not be copied. Members of the UTRGV community uphold the [Vaquero Honor Code](#)'s

shared values of honesty, integrity and mutual respect in our interactions and relationships. In this regard, academic integrity is fundamental in our actions, as any act of dishonesty conflicts as much with academic achievement as with the values of honesty and integrity. The Writing Center is an excellent resource to assist in learning about and avoiding plagiarism in writing. Violations of academic integrity include, but are not limited to: cheating, [plagiarism \(including self-plagiarism\)](#), and collusion; submission for credit of any work or materials that are attributable in whole or in part to another person; taking an examination for another person; any act designed to give unfair advantage to a student; or the attempt to commit such acts (Board of Regents Rules and Regulations, STU 02-100, and UTRGV Academic Integrity Guidelines). **All violations of Academic Integrity will be reported to Student Rights and Responsibilities through [Vaqueros Report It](#).**

Student Support Resources

Student Support Services

Center Name	E-Mail
Advising Center	AcademicAdvising@utrgv.edu
Career Center	CareerCenter@utrgv.edu
Counseling Center	Counseling@utrgv.edu
Food Pantry	FoodPantry@utrgv.edu
Learning Center	LearningCenter@utrgv.edu
University Library	circulation@utrgv.edu
Writing Center	WC@utrgv.edu
UCentral	ucentral@utrgv.edu

Center Name	E-Mail
Brightspace Assistance (COLTT)	Ticket Portal

University Policy Statements

SEXUAL MISCONDUCT AND MANDATORY REPORTING

In accordance with UT System and institutional policies, your instructor is a “Responsible Employee” for mandatory reporting purposes under Title IX regulations and Texas Education Code sections 51.251-252 and must report any instance of sexual misconduct by or against a student or employee of the institution, which includes sexual assault, stalking, dating violence, domestic violence, and sexual harassment to the [Office of Title IX and Equal Opportunity \(otixeo@utrgv.edu\)](#). More information can be found on the [OTIXEO website](#). If students, faculty, or staff would like confidential assistance, or have questions, they can contact OAVP ([Office for Advocacy & Violence Prevention](#)).

STUDENT ACCESSIBILITY SERVICES

[Student Accessibility Services](#) has offices on Brownsville and Edinburg campuses. [Visit the SAS web page to learn more and explore accessibility services.](#)

STUDENTS WITH DISABILITIES

Students with a documented disability (physical, psychological, learning, or other disability which affects academic performance) who would like to receive reasonable academic accommodations should contact [Student Accessibility Services \(SAS\)](#) for additional information. The student must apply for accommodations using the [mySAS portal](#) and is responsible for providing sufficient documentation of the disability to SAS. Upon submission of the request, students should expect to participate in an interactive discussion, or an intake appointment, with SAS staff. Accommodations may be requested at any time but are not granted retroactively, meaning they are valid moving forward after approval by SAS. Academic accommodations will not apply prior to SAS'

approval. Students should contact SAS early in the semester/module for guidance.

Students who experience a broken bone, severe injury, or undergo surgery may also be eligible for temporary accommodations. Please contact [Student Accessibility Services \(SAS\)](#) for more information.

PREGNANCY, PREGNANCY-RELATED, AND PARENTING ACCOMMODATIONS

Title IX of the Education Amendments of 1972 prohibits discrimination based on sex, which includes discrimination based on pregnancy, marital status, or parental status.

Students seeking accommodations related to pregnancy, pregnancy-related condition, or parenting should submit the request using the form found at [Pregnancy and Parenting | UTRGV](#).

MANDATORY COURSE EVALUATION PERIOD

Students have the opportunity to complete an ONLINE evaluation of this course through Watermark Course Evaluations and Surveys, which may be accessed through my.UTRGV or the Brightspace course module. Course evaluations are used by the instructor to inform revisions of the course to ensure student success. Course evaluations are also used by the instructor for annual performance review, promotion applications, teaching award applications, among others.